

## Computer Science 112 - Assignment 1: Writing an Employee Management System

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Submit assignment

If you have a Study.com membership and are seeking college credit for this course, you must submit all projects. Below you will find prompts and instructions for submitting this project.

## About this Project

C++ is a general-purpose object-oriented programming language. The best way to understand programming is to learn about flow, syntax, and debugging through practice. You will create a C++ program that incorporates elements you studied in this course.

AI may be used as a tool to support your process with proper documentation but may **not** create or write any part of your project for you. AI-generated projects will result in an academic integrity violation as outlined in the Study.com Academic Integrity Policy.

# Instructions

For this assignment, you will write a program that keeps track of employees at a company. Three types of users can access the system: Human Resources, Management, and General Employees. A Human Resources user will be able to add, view, search, modify, and delete employees. A Management user can only search and view employees. A General Employee user can only view his or her own file. Employee files will have at least the employee's name and numeric user id.

1. Your program will implement a management information system.
2. The minimum requirements of the system are: implement a base employee class that will be defined for each employee type or function, implement a log-in feature, and the login will determine what functions are accessible to the employee.
3. The program will be interactive. At a minimum, a log-in screen and menu screen will be used. Input should be validated for correct login credentials.
4. Functions and classes should be used where appropriate to produce readable, reusable code. Library functions and user-defined functions should be used where appropriate.
5. The program will be well-commented/documented, including a comment block at the top of the document (document header comment) as well as line comments

Some simple rules to follow to make your code more readable:

- Use indentation or spacing consistently. (e.g., every line in a loop is indented to align)
- Add whitespace to help separate parts of the code. (e.g., empty line after a loop, space after a semicolon in a loop, etc.)
- Give meaningful names to variables. Variables named x, y, and z are not descriptive and give the reader no information about their purpose; instead use names like primary, max, and counter. Loop condition variables are an exception because they are contained to a single area of the code and have a universally known function, so naming with a single letter (i, j, etc.) is acceptable.
- Code should be organized into functions so that segments of code that are reused are placed within functions. Function definitions should be contained in a single section of the program. Functions, like variables, should have meaningful names.

## Formatting

Please submit your project as one of the following:

- a .txt file
- a link to your cloud storage, such as Google Drive, Dropbox, or OneDrive, containing your work. Please ensure the privacy settings allow access. For example, if you're using Google Drive, make sure the file is set to 'anyone with the link can view'.

# Grading

Your project will be graded based on the following rubric:

| Category                       | Unacceptable (0-2)                                                    | Needs Improvement (3-6)                                                                | Good (7-8)                                                               | Excellent (9-10)                                                       |
|--------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| Program Specifications (x5)    | Program does not function consistently or at all.                     | Details of the assignment were violated and/or program exhibits inconsistent behavior. | Assignment is complete and produces correct results in typical cases.    | No errors in any case and checks for exception cases.                  |
| Readability (x2)               | Issues with naming, formatting, and organization.                     | One or two major problems with naming, formatting, or organization.                    | Code is understandable and well-organized.                               | Code is understandable and exceptionally well-organized and formatted. |
| Documentation (x2)             | No comment header or comments in the file                             | File comments header missing or comments missing or not meaningful.                    | Code is appropriately commented but may have a few unnecessary comments. | No errors, code is commented exceptionally well.                       |
| Good Practice/ Efficiency (x1) | Tasks were not done efficiently, succinctly, or using best practices. | Code uses a bad approach in more than one place.                                       | Code uses good practices and is overall efficient.                       | Code is exceptionally designed and efficient.                          |

## Grading Timeline & Policies:

- You should receive your project grade within 48 hours.
- If you are not satisfied with the score you receive, you may revise and resubmit for grading.
- Keep in mind that the grade you receive on your project is only a portion of your overall grade for the course. Please see the course syllabus for a more detailed breakdown of the grading policy.

## Resources

## Related Lessons

- Basic Input & Output in C++ Programming
- Validating Input Data in C++ Programming

- Practical Application for C++ Programming: If Statements
- Practical Application for C++ Programming: Classes
- Standard Library Functions in C++: Definition & Examples

## Tips for Success

### Before You Submit:

- Read the instructions carefully and ensure you've completed all requirements
- Check for errors and add any finishing touches
- Use the rubric provided to self-assess your work

### Avoiding Academic Integrity Violations:

- Review the Study.com Academic Integrity Policy to ensure that you have not violated any rules in the completion of this project.
- Use online plagiarism checkers to make sure that your project is not too similar to any existing materials.
- Use GPTZero to ensure that your project is sufficiently unique and free from AI plagiarism.
- Plagiarized projects will NOT be graded and may result in disciplinary actions.

### Responsible Use of AI:

As outlined in the Study.com Academic Integrity Policy, AI may be used as a tool to support your process for completing this project but may not create or write any part of your project for you unless the specific instructions state AI content generation as a requirement.

- All AI usage for this project must be properly documented using the AI Use Log according to the guidelines in the How to Use AI Tools in College Course Assignments article.
- For additional information on using generative AI tools for learning, please refer to Understanding Generative AI as a Student: Uses, Benefits & Drawbacks.



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